

Introduction to Fuzzy Logic

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Logic

Logic can be used to represent knowledge

Propositional logic: truth values for *propositions*

e.g.: *The grass is green is true*

First order predicate logic: truth values for *predicates*

There are variables and quantifiers

e.g.: $\exists X : (\text{Green } X) \text{ is } \textit{true}$

Second order predicate logic: predicates about predicates

e.g.: $\exists X, Y : (\text{Believe } Y ((\text{Green } X) \text{ is } \textit{true})) \text{ is } \textit{false}$

Propositional logic

Propositional logics are concerned with *propositional* (or *sentential*) operators which may be applied to one or more propositions giving new propositions.

Accent on the *truth value* of propositions, not on their meaning, and on how these truth values are *composed*.

A logic is *truth functional* if the truth value of a compound sentence depends only on the truth values of the constituent atomic sentences, not on their meaning or structure. For such a logic the only important question about propositions is what truth values they may have.

In a *classical*, *boolean* or *two-valued* logic every proposition is either *true* or *false*, and no other feature of the proposition is relevant.

Example propositional logic

For example, the conjunction of the two sentences:

Grass is green

Pigs don't fly

is the sentence:

Grass is green and pigs don't fly

The conjunction of two sentences will be true if, and only if, each of the two sentences from which it was formed is true.

Other operators are disjunction (OR) and negation (NOT)

First order predicate logics

Same as propositional logics augmented with the possibility to define **predicates** on **variables** => existential and universal quantifiers (**Exists**, **For All**)

A **predicate** is a feature of language which you can use to make a statement about something, e.g., to attribute a property to that thing.

If you say "Peter is tall", then you have applied to Peter the predicate "is tall". We also might say that you have predicated tallness of Peter or attributed tallness to Peter.

Inference

In predicate logics it is possible to **infer** the truth value of a proposition by inferential mechanisms, such as **modus ponens**

E.g.: from

All men are mortal (modus ponens)

and

Socrates is a man (proposition)

we can infer that

Socrates is mortal (proposition)

Representational power of classical logic

Aristoteles already had some problems about the validity of "classical" logic as a **knowledge representation** tool.

For instance, how can we state the truth value of a proposition in the future?

E.g.: "Tomorrow it will rain"



Many-valued logics

We may introduce three-valued logics to cope with indecision:
true (1), false (0), and **undefined** (1/2)

So, why not introducing infinite-valued logics:

a continuum of truth values (e.g., in $[0,1]$)

E.g., *Łukasiewicz* (1930) logic L_1

- $T(\neg a) = 1 - T(a)$
- $T(a \wedge b) = \min(T(a), T(b))$
- $T(a \vee b) = \max(T(a), T(b))$
- $T(a \Rightarrow b) = \min(1, 1 + T(b) - T(a))$
- $T(a \Leftrightarrow b) = 1 - |T(a) - T(b)|$

The liar paradox

“I’m a liar”

This sentence is a paradox in classical logic since no formula can have the same truth value of its negation

This may not be so in many-valued logics.

In Łukasiewicz logic, for instance, it can be that the truth value of φ is 0.5, and that of $\neg\varphi$ is the same, so the proposition is consistent with the axioms, and it is no longer a paradox.

Fuzzy logic

Fuzzy logic is an infinite-valued logic, with truth values in $[0,1]$

Propositions are expressed as

A is L,

where:

- A is a linguistic variable
- L is a label denoting a fuzzy set

E.g., Temperature is HIGH

Simple propositions

***p*: X is L**

X is a linguistic variable

L denotes a fuzzy set F, defined on U, which represents a fuzzy predicate

$\mu_F(x)$ is the membership function defining F

$\mu_F(x)$ is interpreted as truth value of the proposition *p*

$$T(p) = \mu_F(x)$$

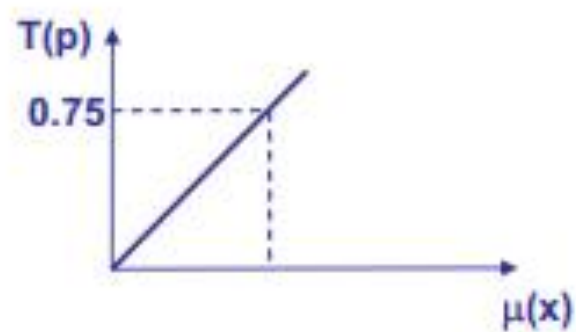
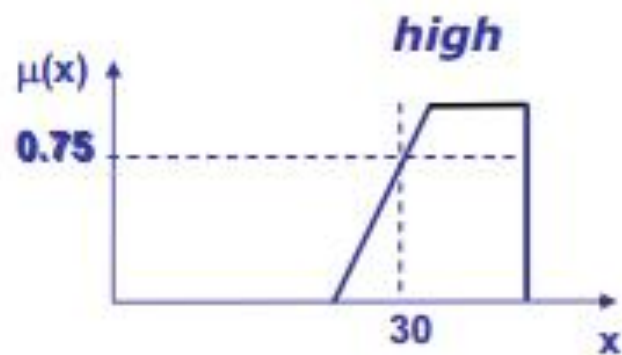
So, the truth value of the proposition P is a fuzzy set defined on [0, 1]

An example

p: Temperature is high

X

L



Degrees of truth and probability

Degrees of truth are often confused with **probabilities**.

They are conceptually distinct: fuzzy truth represents membership to vaguely defined sets, not likelihood of some event or condition.

To illustrate the difference, consider this scenario:

Bob is in a house with two adjacent rooms: the kitchen and the dining room. In many cases, Bob's status within the set of things "in the kitchen" is completely plain: he's either "in the kitchen" or "not in the kitchen". What about when Bob stands in the doorway? He may be considered as "partially in the kitchen". Quantifying this partial state yields a fuzzy set membership. With only his little toe in the dining room, we might say Bob is 0.99 "in the kitchen", for instance. No event (like a coin toss) will resolve Bob to being completely "in the kitchen" or "not in the kitchen", as long as he's standing in that doorway.

Fuzzy modifiers

Fuzzy modifiers modify truth values

x is Young means ***(x is Young) is true***

It can be modified as:

x is very Young is true

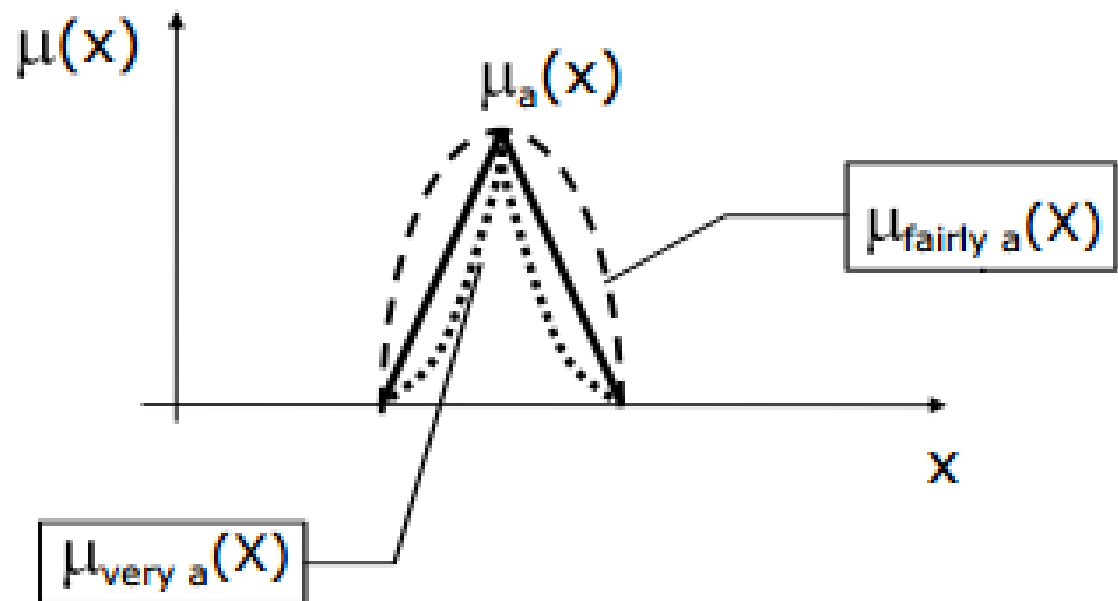
x is Young is very true

x is very Young is very true

Examples of fuzzy modifiers

$$\mu_{\text{very } a}(x) = \mu_a(x)^2$$

$$\mu_{\text{fairly } a}(x) = \mu_a(x)^{1/2}$$



Kind of modifiers

- **Strong modifiers**

$$m(a) \leq a \text{ for each } a \text{ in } [0, 1]$$

They make the predicate stronger, so they reduce the truth of the proposition

- **Weak modifiers**

$$m(a) \geq a \text{ for each } a \text{ in } [0, 1]$$

They make the predicate weaker, so they increase the truth of the proposition

Properties of fuzzy modifiers

They have to satisfy these properties:

- $m(0)=0$ and $m(1)=1$
- m is a continuous function
- if m is strong m^{-1} is weak, and the other way round
- Given another modifier g , the composition of g and m and viceversa are modifiers, too, and ,if both are strong (weak), so is their composition

Qualified, non-conditional propositions

p : (X is F) is S

S is a fuzzy truth qualifier

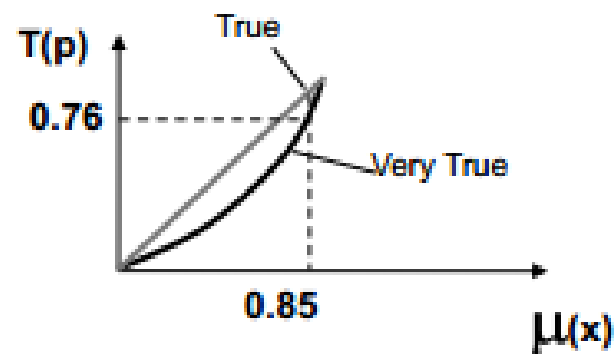
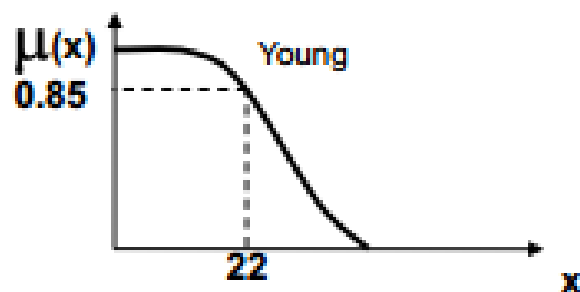
F is a fuzzy set

P is *truth qualified*

Example

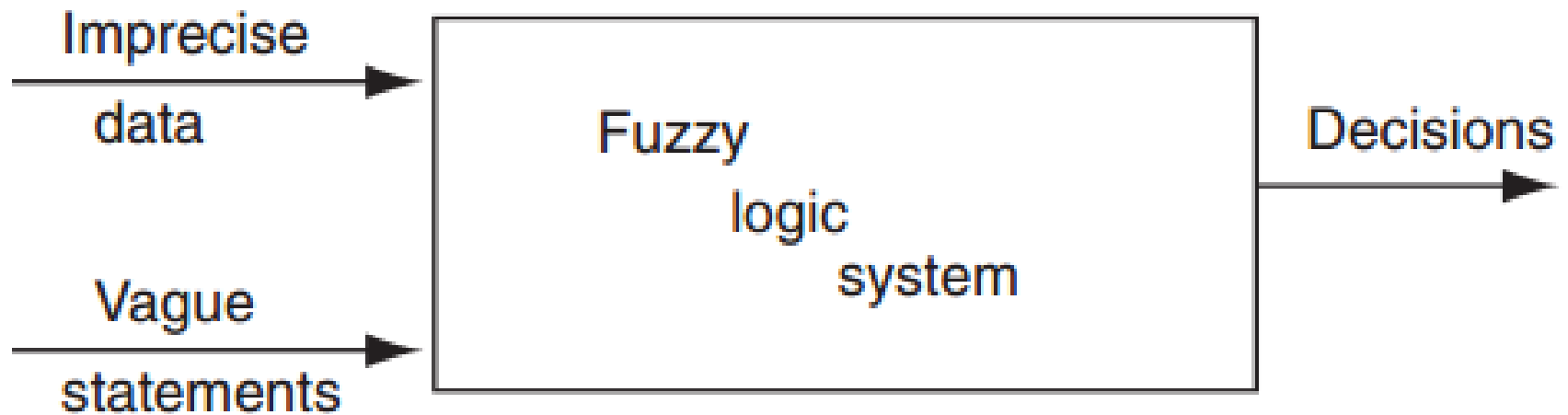
p: Tina is Young is very true

X F S



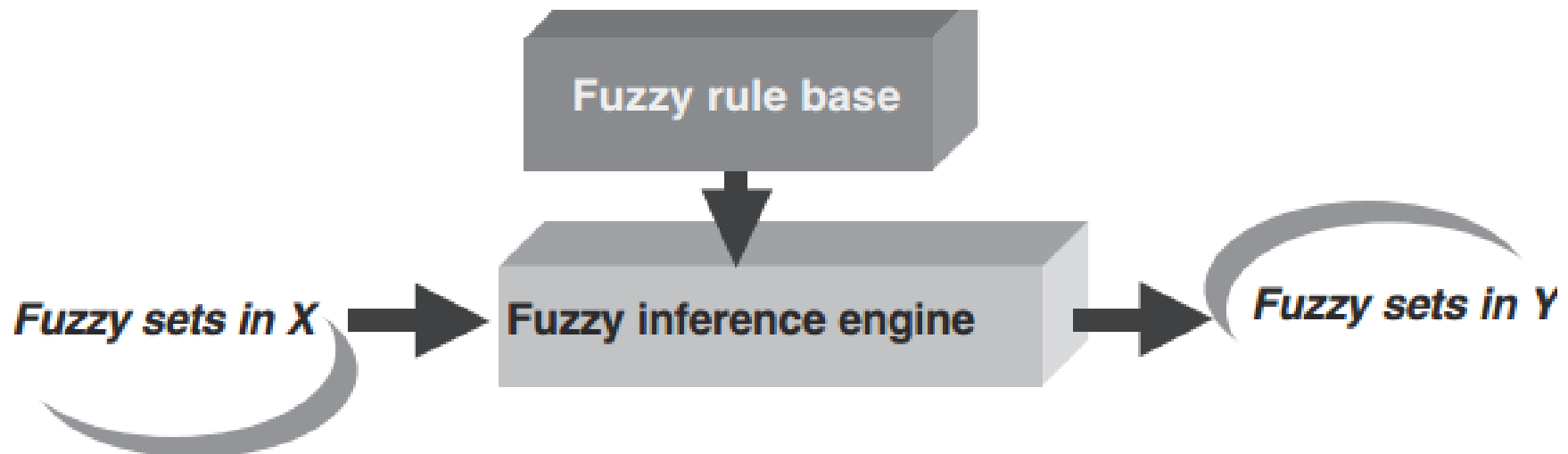
Fuzzy Logic

- ✓ The Fuzzy Logic tool was introduced in 1965
- ✓ A fuzzy logic system which accepts imprecise data and vague statements such as low, medium, high and provides decisions



Configuration of a pure fuzzy system

- ✓ Fuzzy set works as a concept that makes it possible to *treat fuzziness in a quantitative manner*
- ✓ Fuzzy sets form the building blocks for fuzzy **IF–THEN** rules which have the general form “**IF X is A THEN Y is B,**” where **A and B** are fuzzy sets.
- ✓ The term “**fuzzy systems**” refers mostly to systems that are governed by fuzzy **IF– THEN rules**.



Mat LAB – An Overview

- The Matlab environment integrates graphic illustrations with precise numerical calculations, and is a powerful, easy-to-use, and comprehensive tool for performing all kinds of computations and scientific data visualization.
- Matlab has proven to be a very flexible and usable tool for solving problems in many areas.
- Matlab is a high-performance language for technical computing.
- It integrates computation, visualization, and programming in an easy-to-use environment where problems and solutions are expressed in familiar mathematical notation.

Typical use of Mat LAB includes:

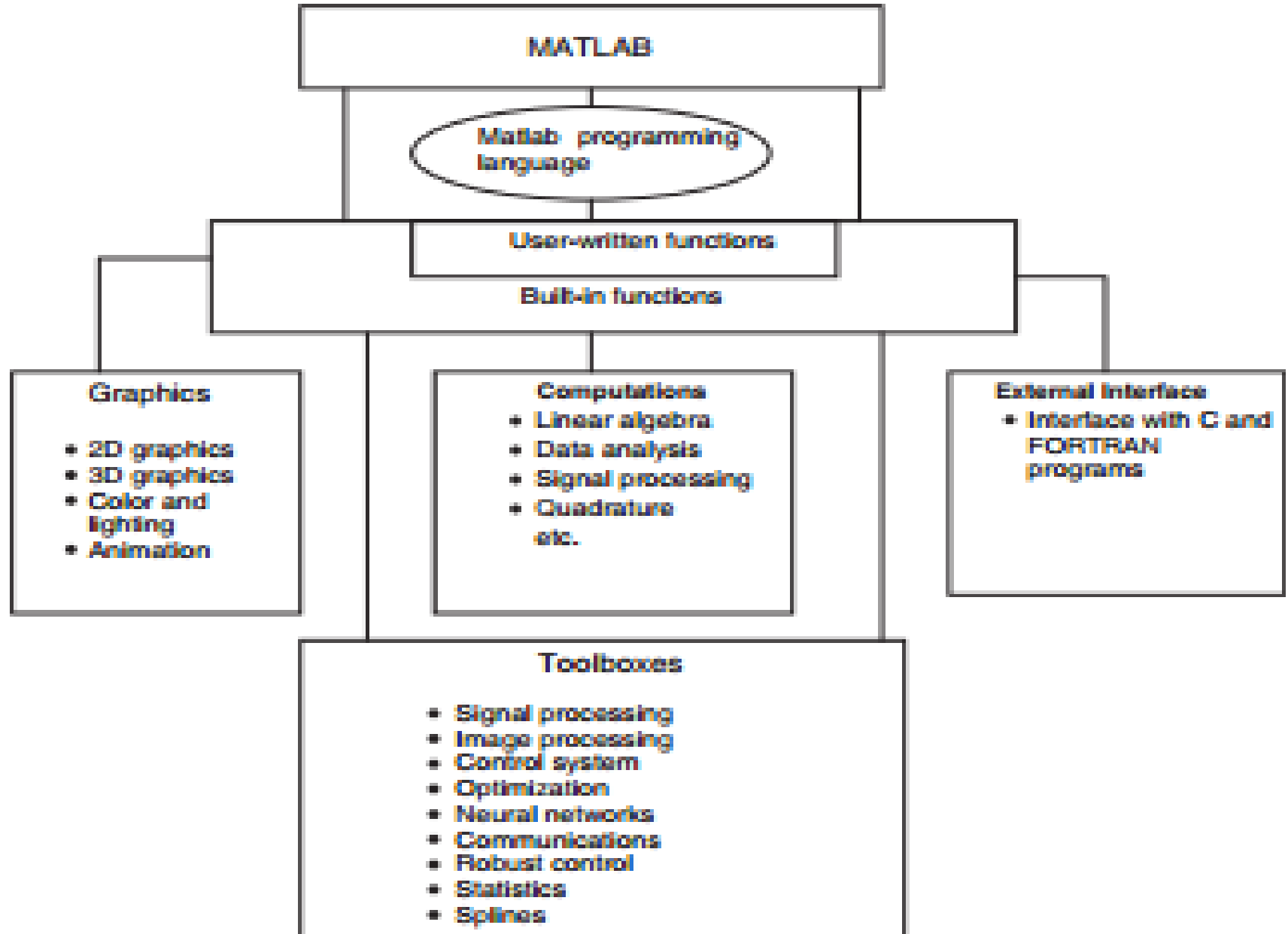
- Math and computation
- Algorithm development
- Modeling, simulation, and prototyping
- Data analysis, exploration, and visualization
- Scientific and engineering graphics
- Application development, including graphical user interface building

.

The main features of Mat LAB are:

- Advance algorithms for high-performance numerical computations, especially in the field of matrix algebra.
- A large collection of predefined mathematical functions and the ability to define one's own functions.
- Two- and three-dimensional graphics for plotting and displaying data.
- A complete help system online.
- Powerful matrix/vector-oriented high-level programming language for individual applications.
- Ability to cooperate with programs written in other languages and for importing and exporting formatted data.
- Toolboxes available for solving advanced problems in several application areas

Shows the main features and capabilities of MatLAB



The toolboxes are:

- ***Control system.*** Provides several features for advanced control system design and analysis
- ***Communications.*** Provides functions to model the components of a communication system's physical layer
- ***Signal processing.*** Contains functions to design analog and digital filters and apply these filters to data and analyze the results
- ***System identification.*** Provides features to build mathematical models of dynamical systems based on observed system data
- ***Robust control.*** Allows users to create robust multivariable feedback control system designs based on the concept of the singular value Bode plot

■

The toolboxes Ctd...:

- ***Simulink.*** Allows you to model dynamic systems graphically
- ***Neural network.*** Allows you to simulate neural networks
- ***Fuzzy logic.*** Allows for manipulation of fuzzy systems and membership functions
- ***Image processing.*** Provides access to a wide variety of functions for reading, writing, and filtering images of various kinds in different ways
- ***Analysis.*** Includes a wide variety of system analysis tools for varying matrices
- ***Optimization.*** Contains basic tools for use in constrained and unconstrained minimization problems

The toolboxes Ctd...:

- ***Spline.*** Can be used to find approximate functional representations of data sets
- ***Symbolic.*** Allows for symbolic (rather than purely numeric) manipulation of functions
- ***User interface utilities.*** Includes tools for creating dialog boxes, menu utilities, and other user interaction for script files

What to remember from these slides?

- What is a fuzzy logic
- The conceptual path from fuzzy sets to fuzzy logic
- Fuzzy modifiers

Questions

